C

EST II — MATH LEVEL 2 • UNIT 5.01

Limits, Continuity & Introductory Derivatives

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	**Q2** A	**Q3** A	**Q4** A	**Q5** A
Q6 A	**Q7** A	**Q8** B	**Q9** C	**Q10** D
Q11 C	**Q12** D	**Q13** D	**Q14** A	**Q15** A
Q16 C	**Q17** B	**Q18** B	**Q19** B	**Q20** A
Q21 B	**Q22** D	**Q23** D	**Q24** A	**Q25** B

Limits, Continuity & Introductory Derivatives Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$ ✓

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

WORKED SOLUTION & VERIFICATION

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$:

$$\frac{7+4i}{2-i} = \frac{(7+4i)(2+i)}{(2-i)(2+i)}$$

Expand the numerator using the distributive property (FOIL):

$$(7+4i)(2+i) = 14 + 7i + 8i + 4i^2$$

Since $i^2 = -1$:

$$14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$$

Expand the denominator using the difference of squares:

$$(2-i)(2+i) = 2^2 - i^2 = 4 - (-1) = 5$$

Now, divide each term in the numerator by the denominator:

$$\frac{10+15i}{5} = \frac{10}{5} + \frac{15i}{5} = 2 + 3i$$

Common Misconceptions:

- Choice B occurs from an arithmetic sign error in the imaginary term.
- Choice C occurs if the student mistakenly calculates the denominator as $2^2 - 1^2 = 3$ by forgetting that $-i^2 = -(-1) = +1$.
- Choice D occurs if the student takes $i^2 = 1$, yielding $14 + 4 = 18$ in the numerator.

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

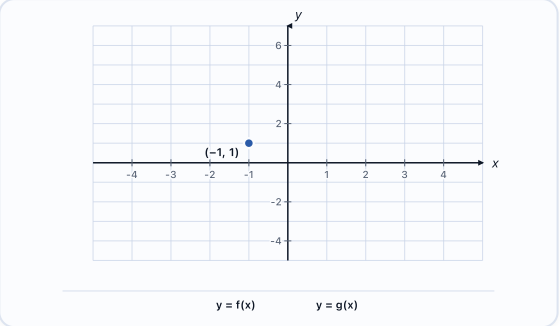
- **A** 0 ✓
- **B** 2
- **C** -1
- **D** 3

WORKED SOLUTION & VERIFICATION

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



- **A** (1, 0) and (3, 0) ✓
- **B** (0, 1) and (0, 3)
- **C** (-1, 0) and (-3, 0)
- **D** (1, 0) and (-3, 0)

WORKED SOLUTION & VERIFICATION

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. \nWe need to solve the quadratic equation $x^2 - 4x + 3 = 0$. \nThis equation can be factored as $(x - 1)(x - 3) = 0$. \nTherefore, the solutions are $x = 1$ and $x = 3$. \nThis means the x-intercepts are $(1, 0)$ and $(3, 0)$.

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

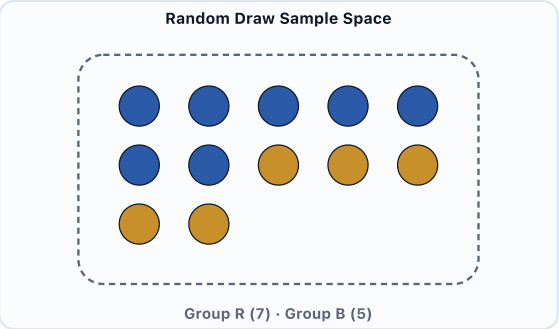
- **A** [3, 4] ✓
- **B** [5, 8]
- **C** [-3, -4]
- **D** [4, 3]

WORKED SOLUTION & VERIFICATION

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



- **A** $\frac{3}{5}$ ✓
- **B** $\frac{2}{5}$
- **C** $\frac{1}{2}$
- **D** $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: $\frac{12 \div 4}{20 \div 4} = \frac{3}{5}$. Thus, the probability of drawing a red marble is $\frac{3}{5}$.

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

- **A** 14 ✓
- **B** 17
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

- **A** 96 ✓
- **B** 192
- **C** 98
- **D** -96

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

B -2 ✓

C 2

D 1

WORKED SOLUTION & VERIFICATION

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$.

Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c):

1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$
2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$
3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$

We can solve this system of equations. Let's subtract equation (1) from equation (2):

$(a + b + c) - (a - b + c) = 1 - 5$

$2b = -4$

$b = -2$

Alternatively, notice that the y-values for $x = -1$ and $x = 3$ are the same (both are 5). This indicates that the axis of symmetry of the parabola is exactly in the middle of $x = -1$ and $x = 3$. The x-coordinate of the vertex (axis of symmetry) is given by $-\frac{b}{2a}$.

Axis of symmetry: $x = \frac{-1+3}{2} = \frac{2}{2} = 1$

So, $-\frac{b}{2a} = 1$, which implies $b = -2a$.

Now, substitute this into equation (2): $a + b + c = 1$

$a + (-2a) + c = 1$

$-a + c = 1 \Rightarrow c = a + 1$

Substitute $b = -2a$ and $c = a + 1$ into equation (1):

$a - (-2a) + (a + 1) = 5$

$a + 2a + a + 1 = 5$

$4a + 1 = 5$

$4a = 4$

$a = 1$

Now we can find b using $b = -2a$:

$b = -2(1)$

$b = -2$

To confirm, we can also find c: $c = a + 1 = 1 + 1 = 2$. The function is $y = x^2 - 2x + 2$.

Let's check the points:

For $(-1, 5)$: $(-1)^2 - 2(-1) + 2 = 1 + 2 + 2 = 5$ (Correct)

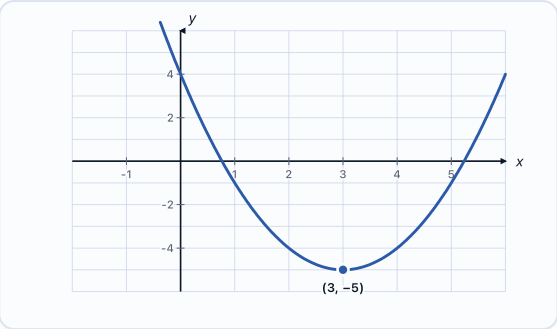
For $(1, 1)$: $(1)^2 - 2(1) + 2 = 1 - 2 + 2 = 1$ (Correct)

For $(3, 5)$: $(3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$ (Correct)

The value of b is -2.

9 MCQ

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



A 9

B -4

C 4 ✓

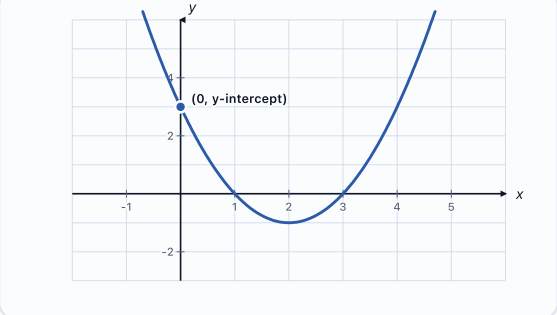
D 1

WORKED SOLUTION & VERIFICATION

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. For $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. The x-coordinate of the vertex is $-(-6)/(2 \cdot 1) = 6/2 = 3$, which matches the given vertex. The y-coordinate of the vertex is $f(3)$. We are given that the vertex is $(3, -5)$, so $f(3) = -5$. Substituting $x = 3$ into the function: $f(3) = (3)^2 - 6(3) + k = 9 - 18 + k = -9 + k$. Setting this equal to the given y-coordinate of the vertex: $-9 + k = -5$. Solving for k: $k = -5 + 9 = 4$. Thus, the value of the parameter k is 4. This means the function is $f(x) = x^2 - 6x + 4$. The vertex y-coordinate calculation is $f(3) = 3^2 - 6(3) + 4 = 9 - 18 + 4 = -5$. The parameter k is 4, not 1. Common algebraic traps include: 1. Errors in the vertex formula. 2. Incorrectly substituting the vertex x-coordinate back into the function. 3. Sign errors when solving for k. In this case, the correct value of k is 4.

10 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



A -3

B 4

C -4

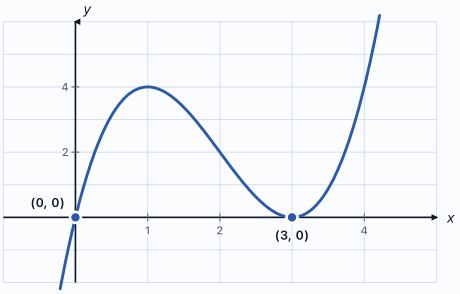
D 3 ✓

WORKED SOLUTION & VERIFICATION

The y-intercept of a function's graph is the point where the graph crosses the y-axis. This occurs when $x = 0$. \nTo find the y-intercept, substitute $x = 0$ into the function: \n$f(0) = (0)^2 - 4(0) + 3$ \n$f(0) = 0 - 0 + 3$ \n$f(0) = 3$ \nTherefore, the y-intercept is 3.

11 MCQ

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.



A 10

B 27/2

C 27/4 ✓

D 9

WORKED SOLUTION & VERIFICATION

First, find the x-intercepts of the function by setting $f(x) = 0$:

$x^3 - 6x^2 + 9x = 0$

$x(x^2 - 6x + 9) = 0$

$x(x-3)^2 = 0$

The x-intercepts are $x = 0$ and $x = 3$ (a repeated root, indicating the graph touches the x-axis at $x = 3$ without crossing).

To find the area, we need to integrate the function between these intercepts. We need to determine if the function is above or below the x-axis in the interval $(0, 3)$. Let's test a point, say $x = 1$:

$f(1) = 1^3 - 6(1)^2 + 9(1) = 1 - 6 + 9 = 4$. Since $f(1) > 0$, the function is above the x-axis in the interval $(0, 3)$.

The area is given by the definite integral:

Area = $\int_0^3 (x^3 - 6x^2 + 9x)dx$

Now, evaluate the integral:

$\int (x^3 - 6x^2 + 9x)dx = \frac{x^4}{4} - 6\frac{x^3}{3} + 9\frac{x^2}{2} = \frac{x^4}{4} - 2x^3 + \frac{9}{2}x^2$

Evaluate the definite integral from 0 to 3:

$[\frac{x^4}{4} - 2x^3 + \frac{9}{2}x^2]_0^3 = (\frac{3^4}{4} - 2(3)^3 + \frac{9}{2}(3)^2) - (\frac{0^4}{4} - 2(0)^3 + \frac{9}{2}(0)^2)$

$= \frac{81}{4} - 2(27) + \frac{9}{2}(9) - 0$

$= \frac{81}{4} - 54 + \frac{81}{2}$

To combine these, find a common denominator, which is 4:

$= \frac{81}{4} - \frac{54\times4}{4} + \frac{81\times2}{4}$

$= \frac{81}{4} - \frac{216}{4} + \frac{162}{4}$

$= \frac{81-216+162}{4}$

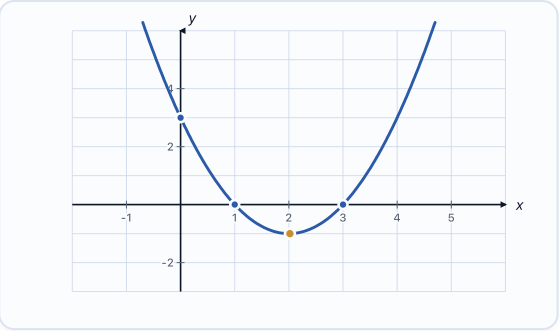
$= \frac{243-216}{4}$

$= \frac{27}{4}$

Thus, the area of the region bounded by the graph of $f(x)$ and the x-axis is $\frac{27}{4}$.

12 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?



A -3

B 0

C -4

D 3 ✓

WORKED SOLUTION & VERIFICATION

The y-intercept of a function occurs where the graph crosses the y-axis, which means the x-coordinate is 0. To find the y-intercept, substitute $x = 0$ into the function $f(x) = x^2 - 4x + 3$. $f(0) = (0)^2 - 4(0) + 3$

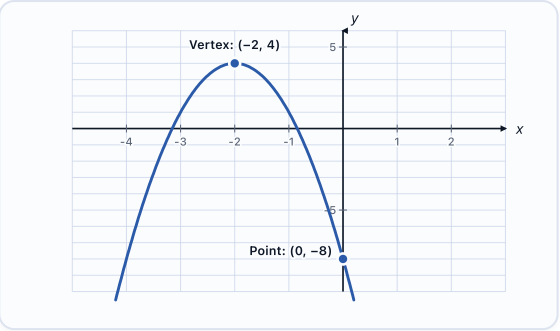
$f(0) = 0 - 0 + 3$

$f(0) = 3$

Therefore, the y-intercept is 3. Common errors might arise from incorrectly substituting values or misinterpreting what the y-intercept represents.

13 MCQ

Consider the function $g(x) = ax^2 + bx + c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2, 4)$. The parabola passes through the point $(0, -8)$. Find the value of the parameter a.



A 2

B 4

C -4

D -2 ✓

WORKED SOLUTION & VERIFICATION

The general form of a quadratic function is $g(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $g(x) = a(x - h)^2 + k$, where (h, k) is the vertex.

Given the vertex is $(-2, 4)$, we can write the function as $g(x) = a(x - (-2))^2 + 4$, which simplifies to $g(x) = a(x + 2)^2 + 4$.

We are also given that the parabola passes through the point $(0, -8)$. We can substitute these coordinates into the vertex form of the equation to solve for a.

$-8 = a(0 + 2)^2 + 4$

$-8 = a(2)^2 + 4$

$-8 = 4a + 4$

Subtract 4 from both sides:

$-8 - 4 = 4a$

$-12 = 4a$

Divide by 4:

$a = \frac{-12}{4}$

$a = -3$

Common Trap: A common mistake is to assume the parabola passes through the origin or to confuse the vertex coordinates with other points. Another mistake is to incorrectly expand the squared term or solve the linear equation for 'a'. For example, if one were to incorrectly use $a = a(x - h)^2 + k$ and substitute $x = 0$ and $y = -8$ into the expanded form of the general quadratic without first establishing the relationship between the general coefficients and the vertex form, they might arrive at an incorrect value for a. For instance, if one incorrectly assumes $c = -8$ and then tries to relate b to the vertex's x-coordinate using $-b/(2a) = -2$ without considering the vertex's y-coordinate, they might get stuck or derive an incorrect 'a'.

14 MCQ

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

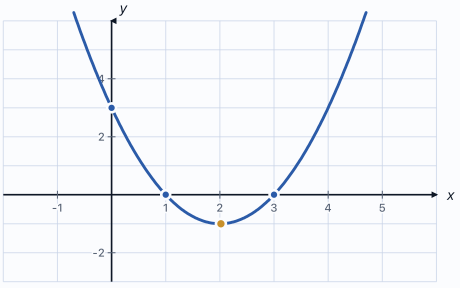
- **A** 4 ✓
- **B** 6
- **C** 5
- **D** 3

WORKED SOLUTION & VERIFICATION

The function is $f(x) = x^3 - 6x^2 + 5$. The derivative is $f'(x) = 3x^2 - 12x$. At $x = a = 1$, the slope of the tangent line is $m = f'(1) = 3(1)^2 - 12(1) = 3 - 12 = -9$. The y-coordinate of point P is $f(1) = 1^3 - 6(1)^2 + 5 = 1 - 6 + 5 = 0$. So, $P = (1, 0)$. The equation of the tangent line at P is $y - 0 = -9(x - 1)$, which simplifies to $y = -9x + 9$.
To find the intersection points of the tangent line and the curve, we set the function equal to the tangent line equation: $x^3 - 6x^2 + 5 = -9x + 9$. Rearranging this equation gives $x^3 - 6x^2 + 9x - 4 = 0$. We know that $x = 1$ is a root because it is the point of tangency, meaning it is a repeated root (at least a double root). Therefore, $(x - 1)^2$ must be a factor of the polynomial. Let's perform polynomial division or synthetic division. Since $x = 1$ is a root, $(x - 1)$ is a factor. $x^3 - 6x^2 + 9x - 4 = (x - 1)(x^2 - 5x + 4)$. Factoring the quadratic term, we get $x^2 - 5x + 4 = (x - 1)(x - 4)$. Thus, the equation becomes $(x - 1)(x - 1)(x - 4) = 0$, or $(x - 1)^2(x - 4) = 0$. The roots are $x = 1$ (a double root) and $x = 4$. The point Q is the other intersection point, so its x-coordinate is $b = 4$.

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Find the x-coordinate of the vertex of the parabola.



- **A** 2 ✓
- **B** 1
- **C** 3
- **D** -1

WORKED SOLUTION & VERIFICATION

The given function is a quadratic function of the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Therefore, the x-coordinate of the vertex is 2.

16 MCQ

Find $\lim_{x \to -3} (-x^2 - 3x - 3)$.

A -5	B 3
C **-3** ✓	D -2

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = -3$ to get -3.

17 MCQ

Find $\lim_{x \to -3} (4x^2 + 4x - 5)$.

A 17	B **19** ✓
C -19	D 20

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = -3$ to get 19.

18 MCQ

Find $\lim_{x \to 3} (-x - 2)$.

A -4	B **-5** ✓
C -7	D 5

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 3$ to get -5.

19 MCQ

Find $\lim_{x \to -5} (2x + 6)$.

A -3	B **-4** ✓
C -6	D 4

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = -5$ to get -4.

20 MCQ

Find $\lim_{x \to 2} (-2x^2 + 2x - 7)$.

- **A** -11 ✓
- **B** 11
- **C** -10
- **D** -13

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 2$ to get -11.

21 MCQ

Find $\lim_{x \to 5} (3x^2 - 7x - 5)$.

- **A** 36
- **B** 35 ✓
- **C** 33
- **D** -35

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 5$ to get 35.

22 MCQ

Find $\lim_{x \to 4} (-3x^2 - 6x + 2)$.

- **A** -69
- **B** -72
- **C** 70
- **D** -70 ✓

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 4$ to get -70.

23 MCQ

Find $\lim_{x \to 0} (2x^2 + 2x)$.

- **A** -2
- **B** 4
- **C** 1
- **D** 0 ✓

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 0$ to get 0.

24 MCQ

Calculate $\lim_{x \to 3} (-2x^2 + 5x - 5)$.

- **A** -8 ✓
- **B** -7
- **C** -10
- **D** 8

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = 3$ to get -8.

25 MCQ

Find $\lim_{x \to -1} (-4x^2 + x - 1)$.

- **A** -5
- **B** -6 ✓
- **C** -8
- **D** 6

WORKED SOLUTION & VERIFICATION

Direct substitution: plug in $x = -1$ to get -6.